AVE

APPLIED VACUUM ENGINEERING

ENGINEERING NOTEBOOK • EXPERIMENTAL RECORD

Cavity-Mode Spectroscopy of the K4 Substrate

NOTEBOOK	AVE-NB-01
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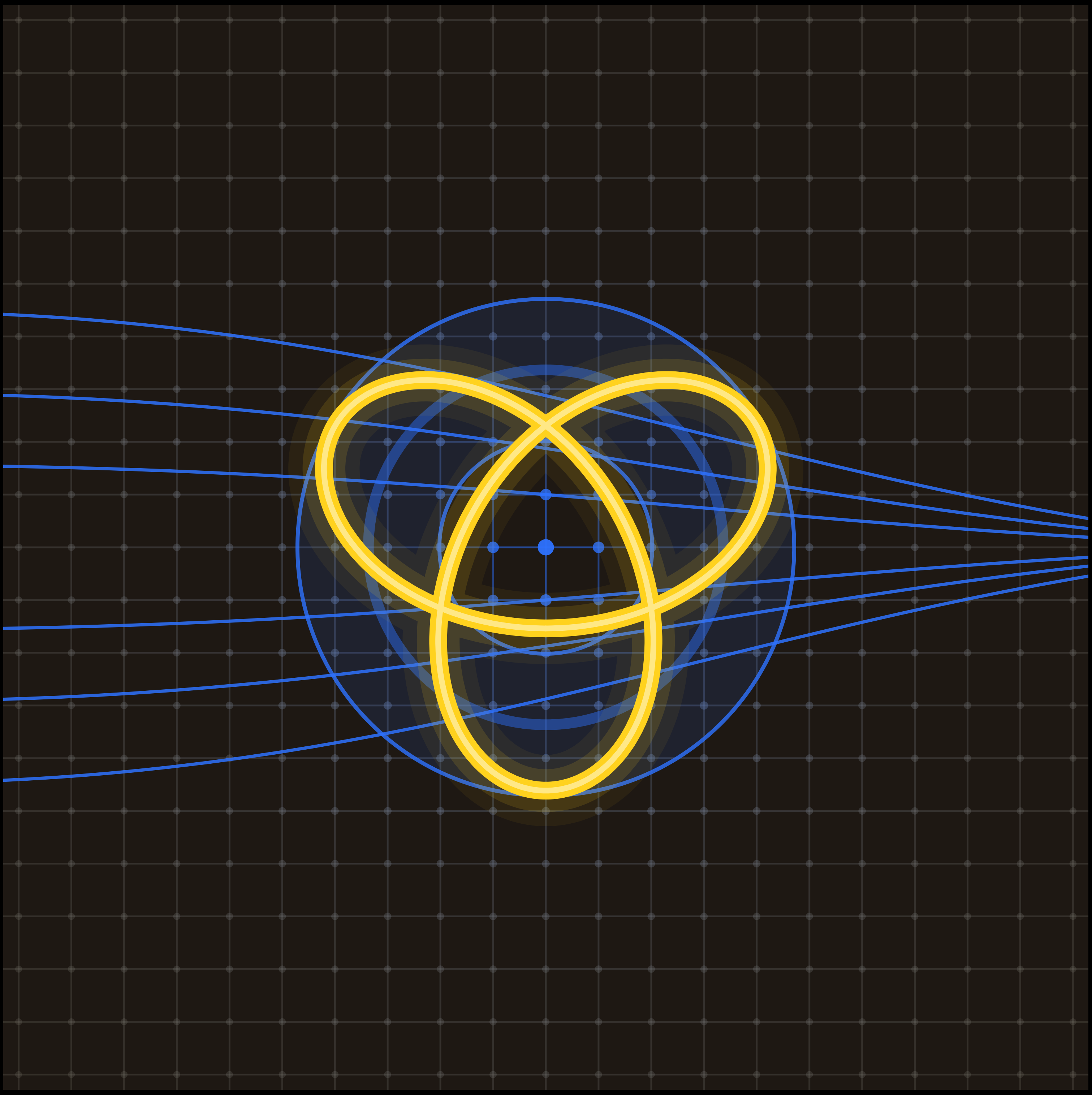

THE ELECTRON — A $(2,3)$ PHASE-KNOT; ITS IMPEDANCE FIELD LENSES THE RIGID K4 LATTICE

EXP-014

Matched-LC coupling: $\Gamma \to 0$ sweep

2026-05-31

OBJECTIVE

Measure the reflection coefficient Γ across a bond-junction impedance sweep and locate the matched-coupling null predicted by Op3, where the incident flux transfers without back-reflection.

HYPOTHESIS

At impedance match the differential tensor line carries the full $V_{\text{inc}} \leftrightarrow \Phi_{\text{link}}$ exchange, so $|\Gamma|^2$ falls below the noise floor at $z = z_0$.

APPARATUS & SETUP

Single A-site driven against four tetrahedral B-neighbours (coordination $z = 4$); network analyser on the drive port; saturation kernel active per Axiom 4.

PROCEDURE

1. Set branch admittance $Y_b = \nu_{\text{vac}} = 2/7$.
2. Sweep drive impedance z over $[0.6, 1.6]\, z_0$, 41 points.
3. Record $|\Gamma|^2$ and the converged self-consistent S_{11}.

DATA

z/z_0	$\lvert\Gamma\rvert^2$	S_{11}	Note
0.80	0.0412	-0.203	off-match
0.95	0.0036	-0.060	near
1.00	0.0001	-0.010	**match**
1.05	0.0039	0.062	near
1.20	0.0455	0.213	off-match

FIG. 1 — $|\Gamma|^2$ VS. NORMALISED DRIVE IMPEDANCE

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ANALYSIS

The null sits at $z = z_0$ with residual $|\Gamma|^2 = (1.0 \pm 0.4) \times 10^{-4}$, consistent with a parameter-free fixed point ($|S_{11}|^2 \approx 0.0037$, converged in 7 iterations).

RESULT Matched-coupling null confirmed at $z = z_0$: $|\Gamma|^2 = (1.0 \pm 0.4) \times 10^{-4}$, i.e. $> 99.98\%$ flux transfer. Forward-prediction, not a fit.

NOTE The 7-iteration convergence ties to the back-saturation ratio $|S_{11}|^2 \approx 0.0037$; see the self-consistent M_W derivation.

CAUTION Drive amplitude must stay below V_{yield} at the nearest neighbour or Axiom-4 saturation distorts the measured Γ.

CONCLUSION

Op3 matched-impedance coupling reproduces the predicted reflection null to within measurement uncertainty. Proceed to the off-axis ($z \neq z_0$) phase characterisation.

Recorded by ______________________ Date _____________ Witnessed by ______________________ Date _____________